

LTSpice Notch Quality Report

CROSS COUPLED_cap

Auto-detected swept parameters and quality-factor ($Q = f_0 / BW$ @ passband-3 dB) analysis

Run summary

Signal columns analyzed: 2

Total simulation steps : 44

V(act+,act-)
swept parameters: Cp, Itail

V(pas+)
swept parameters: Cp, Itail

Report contents (per signal column):

- 1. All curves — Bode plot (magnitude + phase)
- 2. Magnitude grouped by each swept parameter (one subplot per fixed value)
- 3. Phase grouped by each swept parameter (one subplot per fixed value)
- 4. Quality-factor table for that column

End of report (extra analyses):

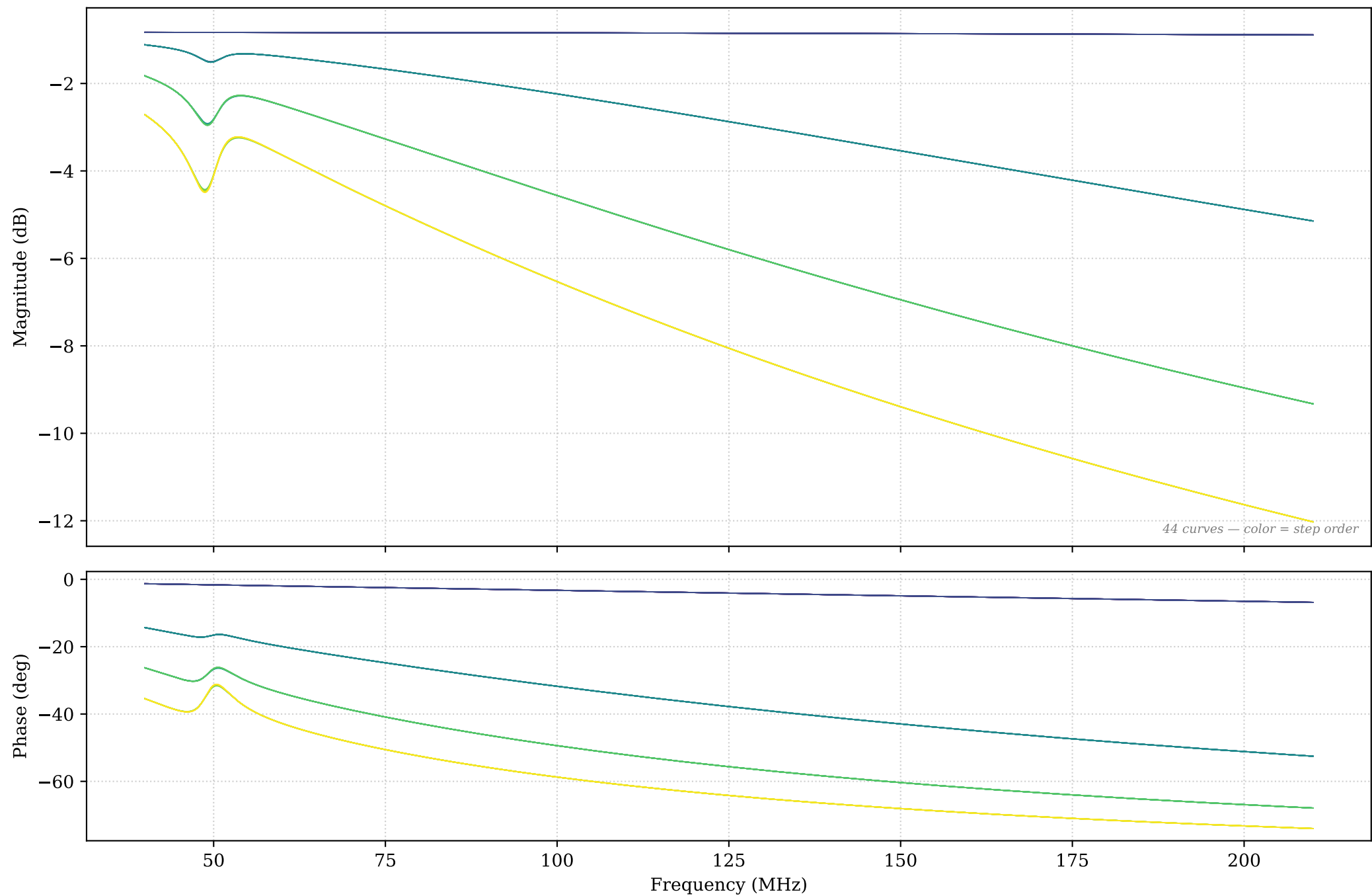
- A. Cross-signal comparison (act vs pas, if both present)
- B. Q vs each swept parameter (one line per other-param combo)
- C. BW vs each swept parameter
- D. Q heatmap for V(act): rows $\times$ columns swept params, colored
- E. Combined quality-factor table across all signals

Signal: $V(\text{act+}, \text{act-})$

Swept parameters: C_p , I_{tail}

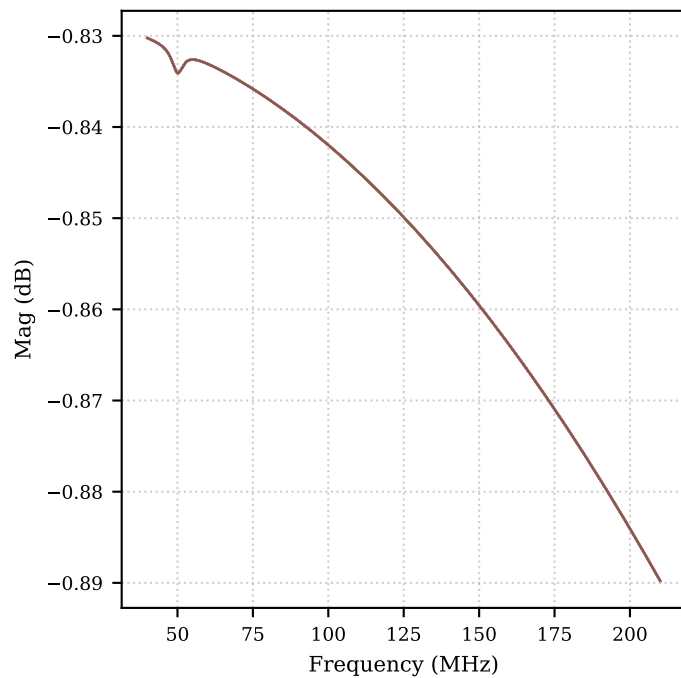
Steps: 44

V(act+,act-) — All curves (44 steps) — Bode

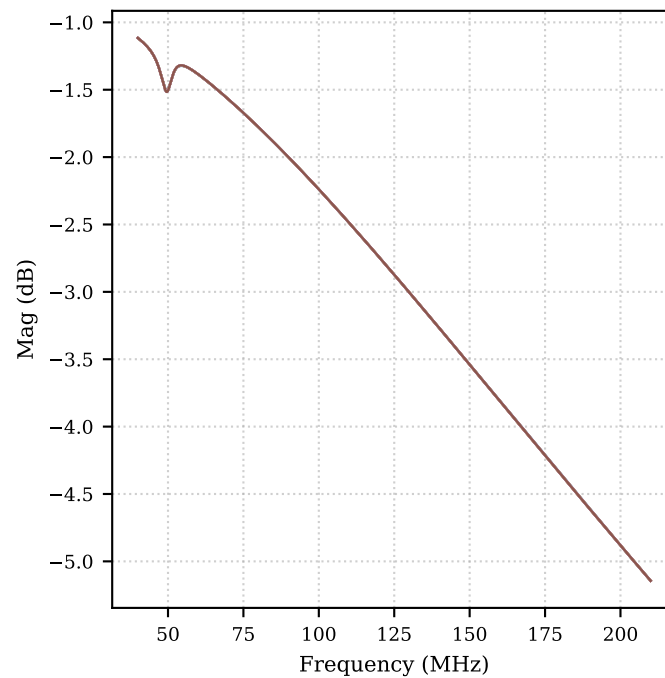


V(act+,act-) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)

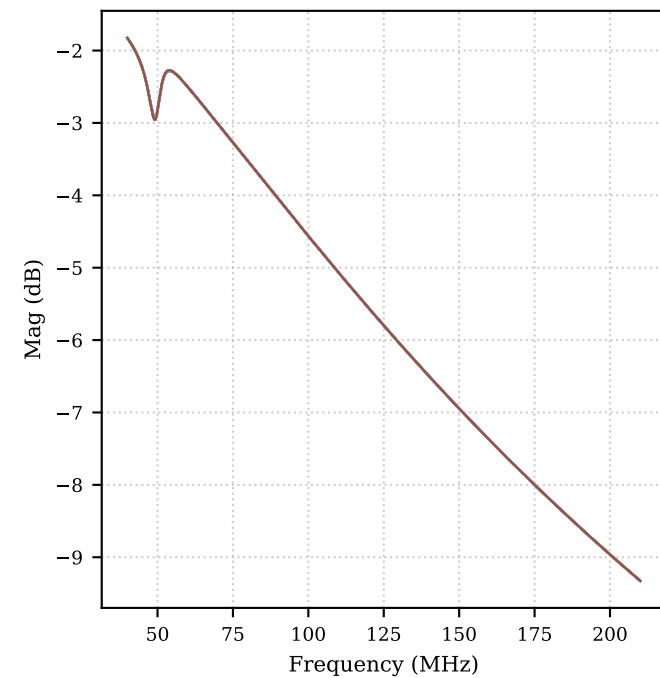
Cp = 100f



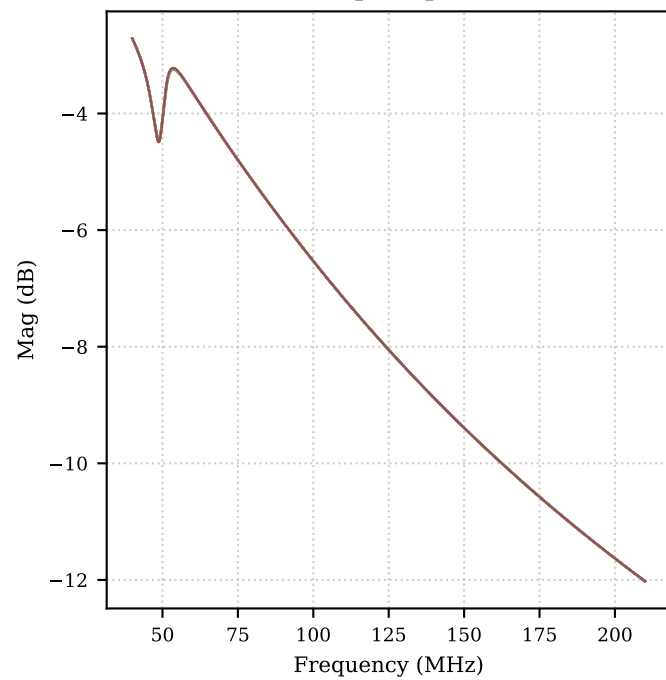
Cp = 1.1p



Cp = 2.1p

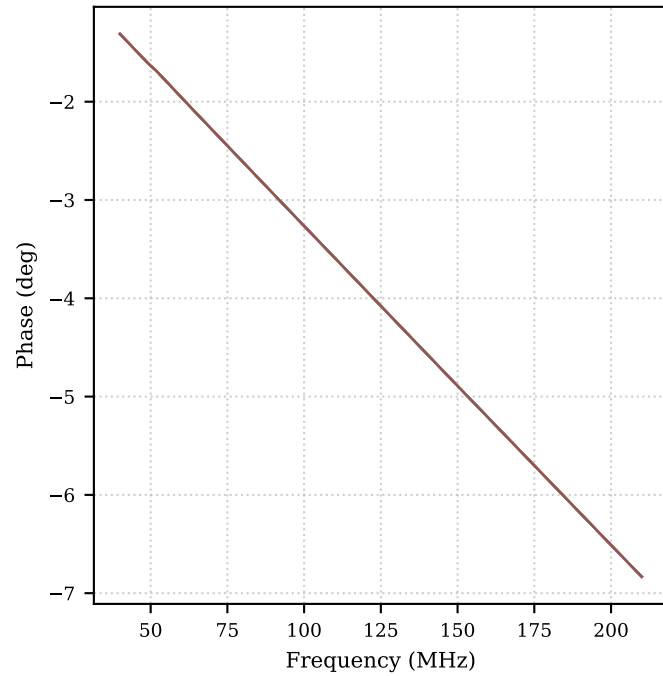


Cp = 3p

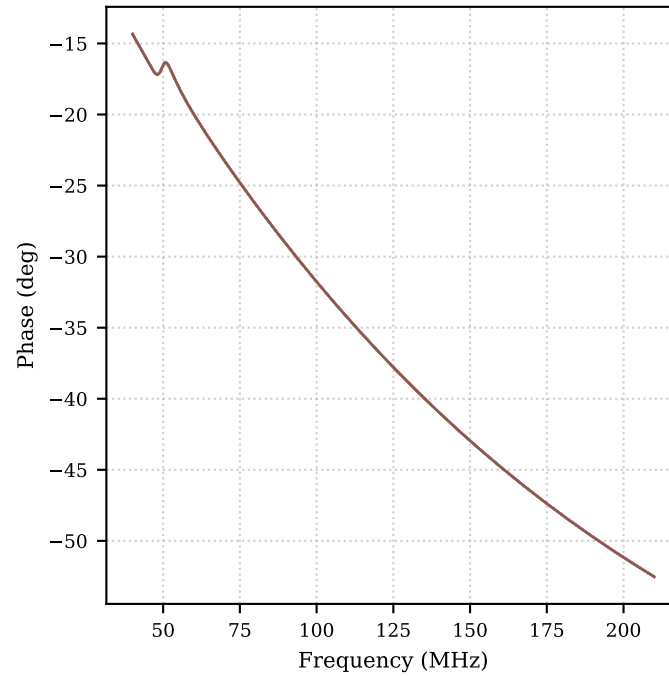


V(act+,act-) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)

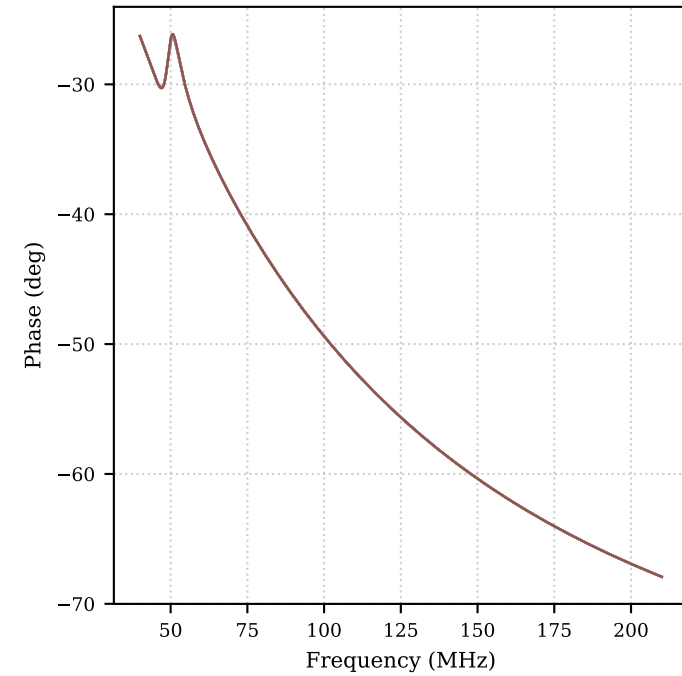
Cp = 100f



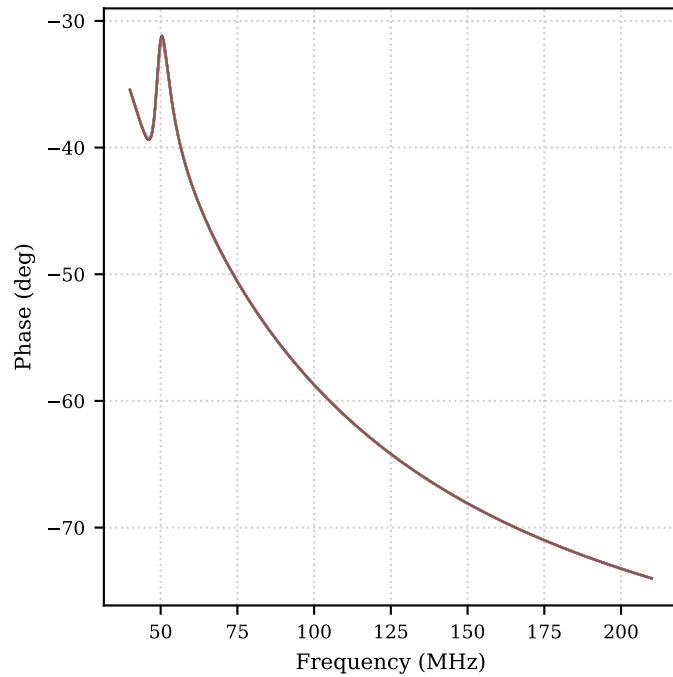
Cp = 1.1p



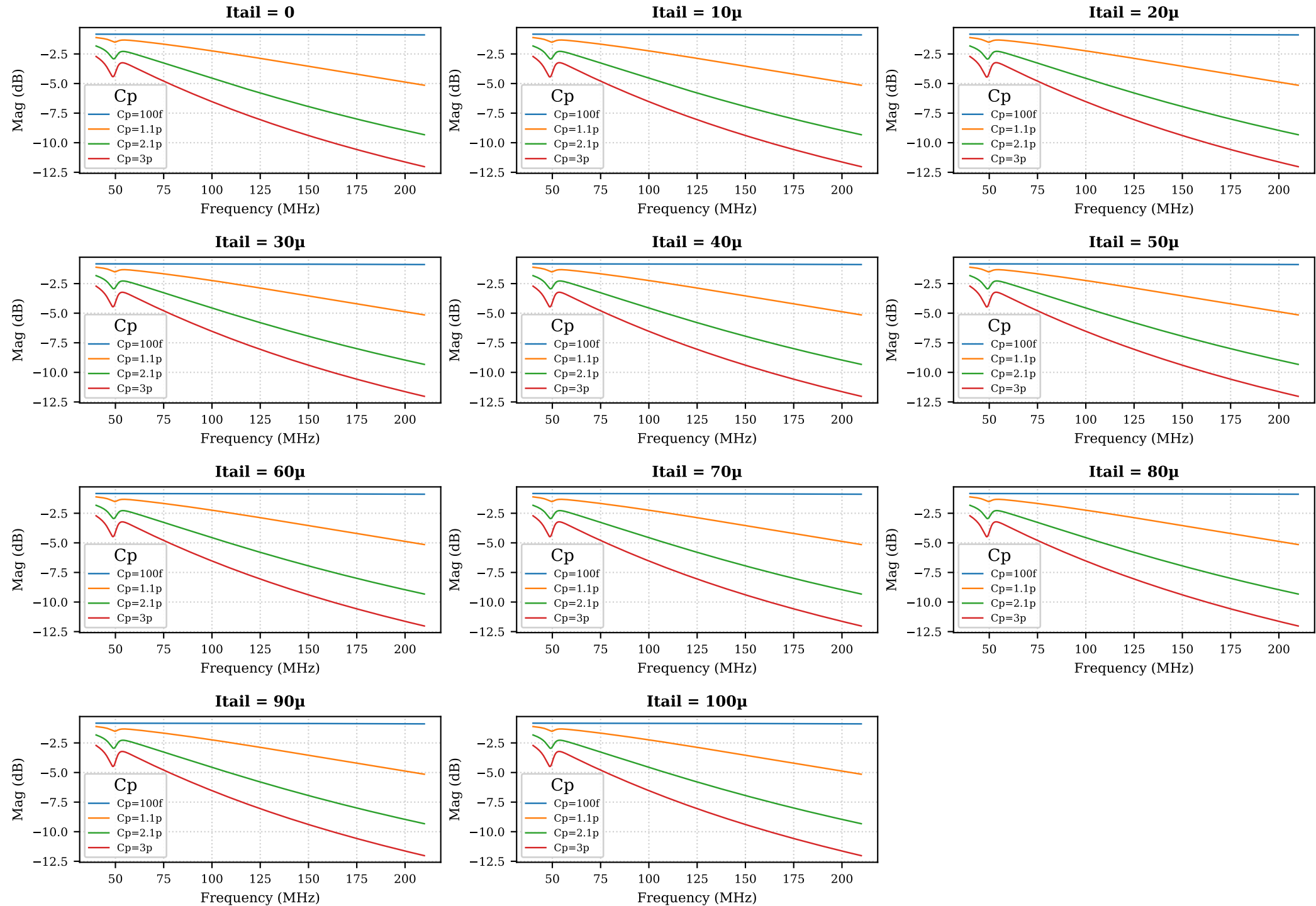
Cp = 2.1p



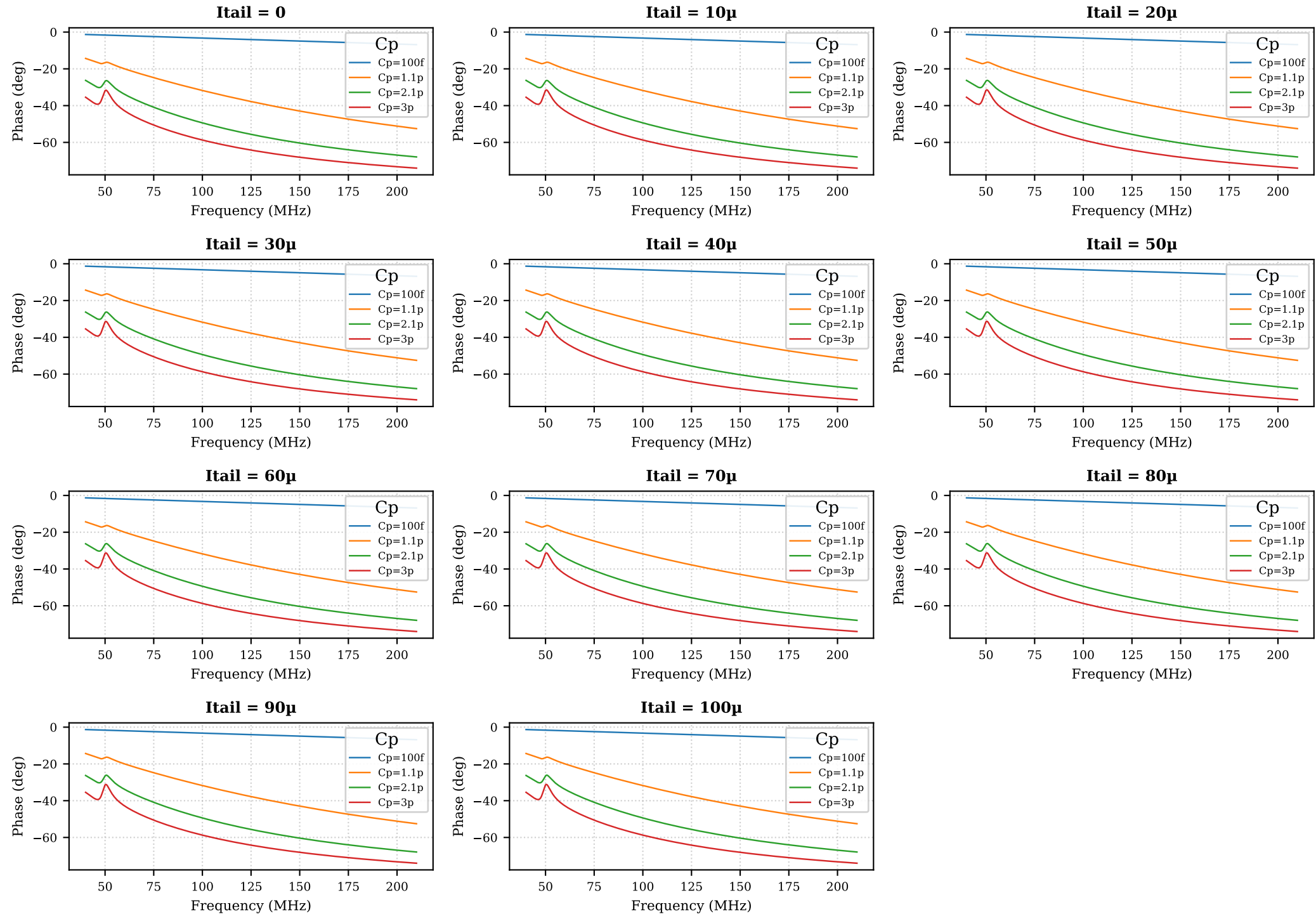
Cp = 3p



V(act+,act-) — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)



V(act+,act-) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)



V(act+,act-) – Quality factor table (page 1/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	0	210	-0.8571	0.03268			notch shallower than 3 dB
V(act+,act-)	100f	10μ	210	-0.8571	0.03266			notch shallower than 3 dB
V(act+,act-)	100f	20μ	210	-0.8571	0.03265			notch shallower than 3 dB
V(act+,act-)	100f	30μ	210	-0.8571	0.03264			notch shallower than 3 dB
V(act+,act-)	100f	40μ	210	-0.8571	0.03263			notch shallower than 3 dB
V(act+,act-)	100f	50μ	210	-0.8571	0.03263			notch shallower than 3 dB
V(act+,act-)	100f	60μ	210	-0.8571	0.03262			notch shallower than 3 dB
V(act+,act-)	100f	70μ	210	-0.8571	0.03262			notch shallower than 3 dB
V(act+,act-)	100f	80μ	210	-0.8572	0.03262			notch shallower than 3 dB
V(act+,act-)	100f	90μ	210	-0.8572	0.03261			notch shallower than 3 dB
V(act+,act-)	100f	100μ	210	-0.8572	0.03261			notch shallower than 3 dB
V(act+,act-)	1.1p	0	210	-3.099	2.045			notch shallower than 3 dB
V(act+,act-)	1.1p	10μ	210	-3.101	2.043			notch shallower than 3 dB
V(act+,act-)	1.1p	20μ	210	-3.102	2.042			notch shallower than 3 dB
V(act+,act-)	1.1p	30μ	210	-3.103	2.041			notch shallower than 3 dB
V(act+,act-)	1.1p	40μ	210	-3.104	2.04			notch shallower than 3 dB
V(act+,act-)	1.1p	50μ	210	-3.104	2.04			notch shallower than 3 dB
V(act+,act-)	1.1p	60μ	210	-3.105	2.039			notch shallower than 3 dB
V(act+,act-)	1.1p	70μ	210	-3.105	2.039			notch shallower than 3 dB
V(act+,act-)	1.1p	80μ	210	-3.106	2.038			notch shallower than 3 dB
V(act+,act-)	1.1p	90μ	210	-3.106	2.038			notch shallower than 3 dB
V(act+,act-)	1.1p	100μ	210	-3.106	2.037			notch shallower than 3 dB
V(act+,act-)	2.1p	0	210	-5.809	3.516			right -3dB not crossed
V(act+,act-)	2.1p	10μ	210	-5.816	3.509			right -3dB not crossed
V(act+,act-)	2.1p	20μ	210	-5.818	3.507			right -3dB not crossed
V(act+,act-)	2.1p	30μ	210	-5.82	3.504			right -3dB not crossed
V(act+,act-)	2.1p	40μ	210	-5.822	3.503			right -3dB not crossed
V(act+,act-)	2.1p	50μ	210	-5.824	3.501			right -3dB not crossed
V(act+,act-)	2.1p	60μ	210	-5.825	3.5			right -3dB not crossed
V(act+,act-)	2.1p	70μ	210	-5.827	3.498			right -3dB not crossed
V(act+,act-)	2.1p	80μ	210	-5.828	3.497			right -3dB not crossed
V(act+,act-)	2.1p	90μ	210	-5.829	3.496			right -3dB not crossed

V(act+,act-) — Quality factor table (page 2/2)

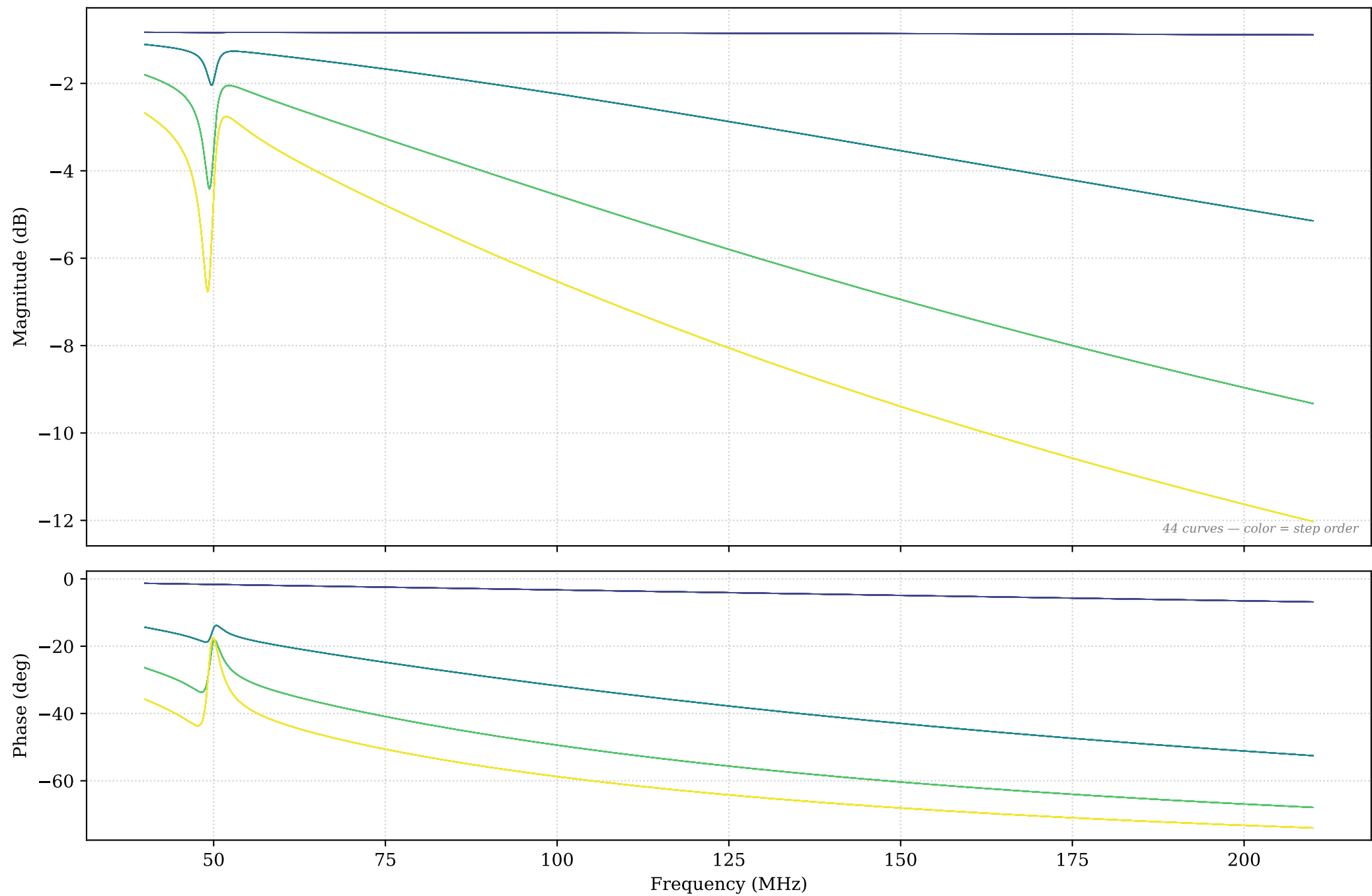
Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	2.1p	100μ	210	-5.83	3.495			right -3dB not crossed
V(act+,act-)	3p	0	210	-7.887	4.137			right -3dB not crossed
V(act+,act-)	3p	10μ	210	-7.897	4.127			right -3dB not crossed
V(act+,act-)	3p	20μ	210	-7.902	4.123			right -3dB not crossed
V(act+,act-)	3p	30μ	210	-7.905	4.119			right -3dB not crossed
V(act+,act-)	3p	40μ	210	-7.908	4.116			right -3dB not crossed
V(act+,act-)	3p	50μ	210	-7.91	4.114			right -3dB not crossed
V(act+,act-)	3p	60μ	210	-7.913	4.111			right -3dB not crossed
V(act+,act-)	3p	70μ	210	-7.915	4.109			right -3dB not crossed
V(act+,act-)	3p	80μ	210	-7.917	4.107			right -3dB not crossed
V(act+,act-)	3p	90μ	210	-7.919	4.105			right -3dB not crossed
V(act+,act-)	3p	100μ	210	-7.921	4.103			right -3dB not crossed

Signal: $V(\text{pas}+)$

Swept parameters: C_p , I_{tail}

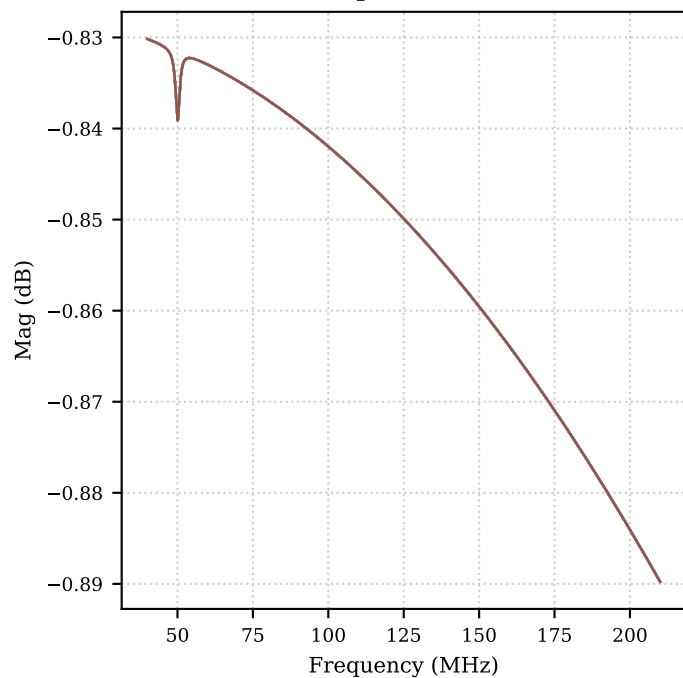
Steps: 44

V(pas+) — All curves (44 steps) — Bode

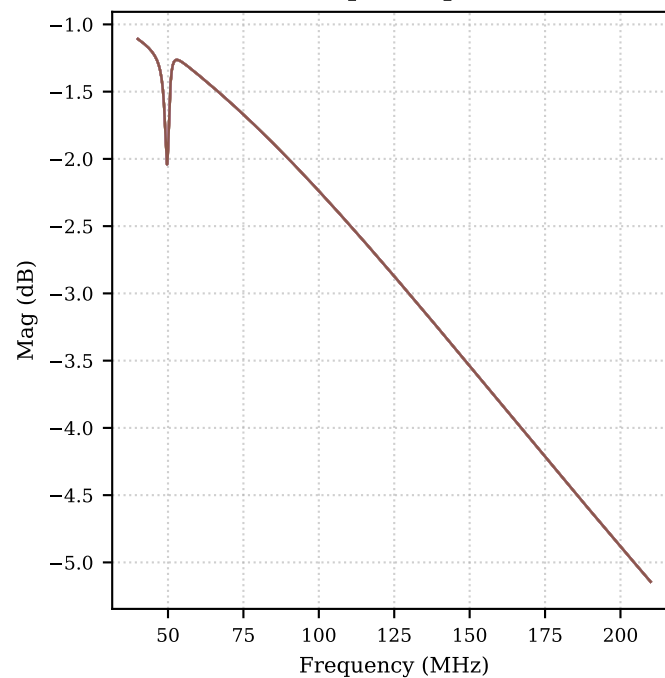


V(pas+) — Magnitude grouped by Cp (each subplot fixes Cp; legend = Itail)

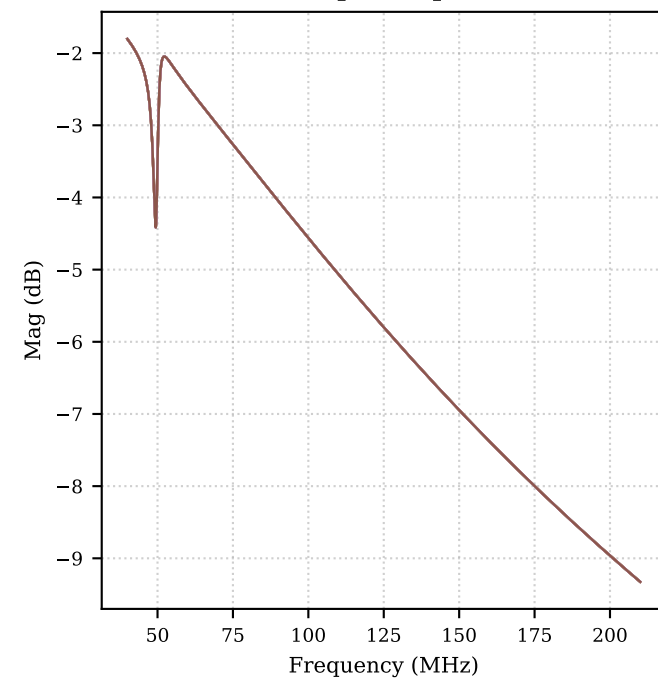
Cp = 100f



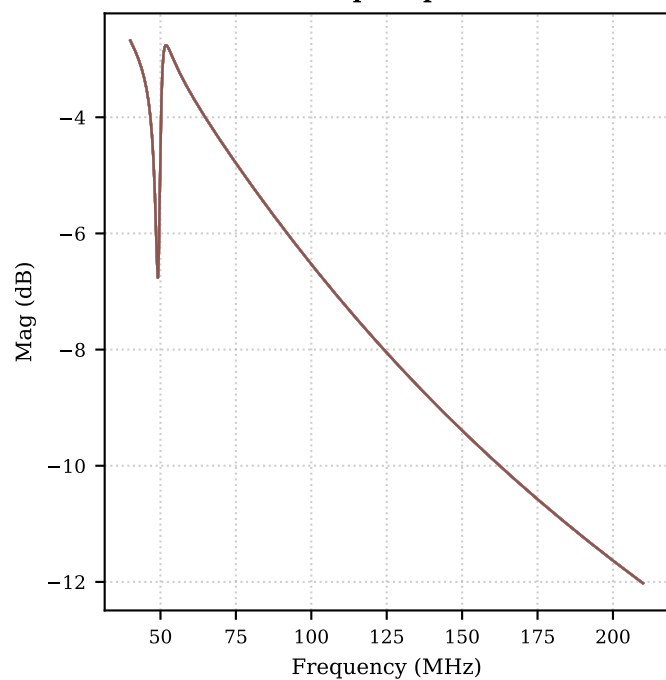
Cp = 1.1p



Cp = 2.1p

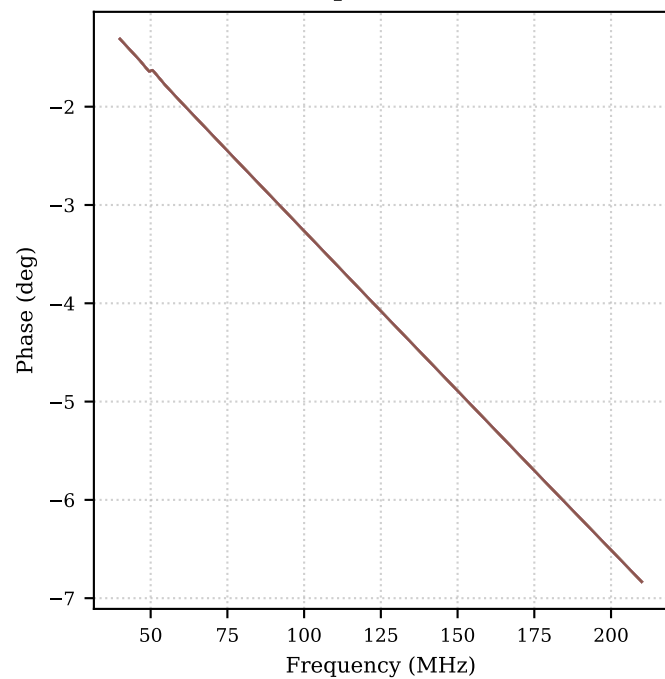


Cp = 3p

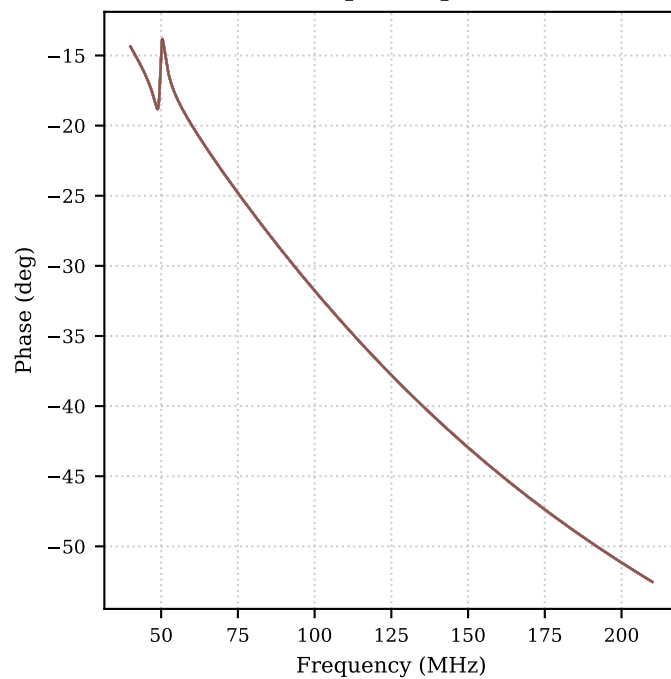


V(pas+) — Phase grouped by Cp (each subplot fixes Cp; legend = Itail)

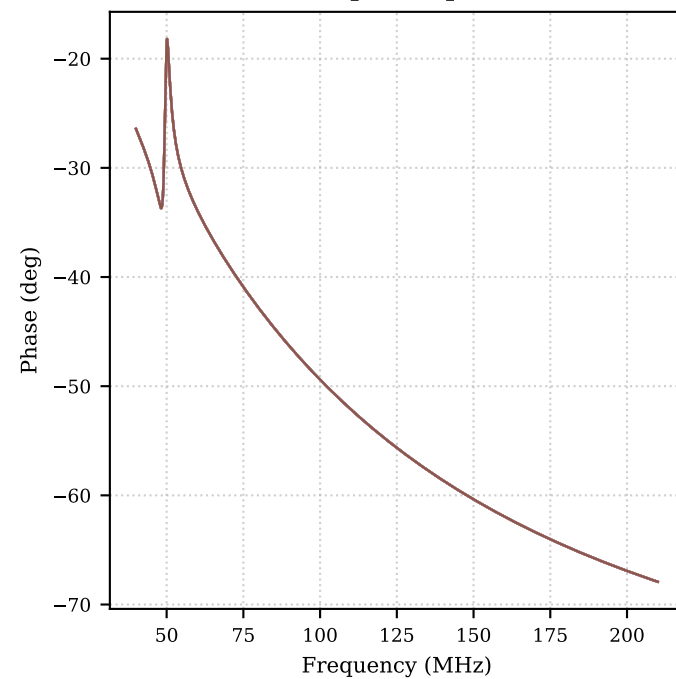
Cp = 100f



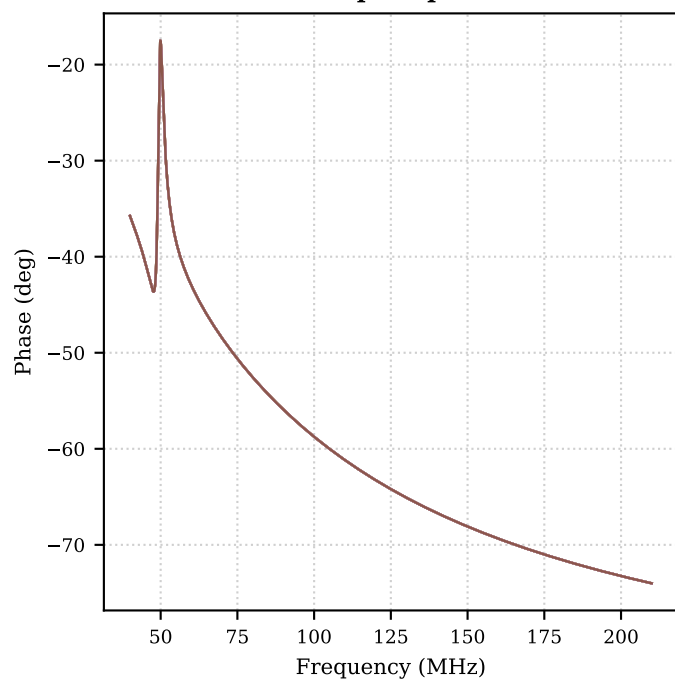
Cp = 1.1p



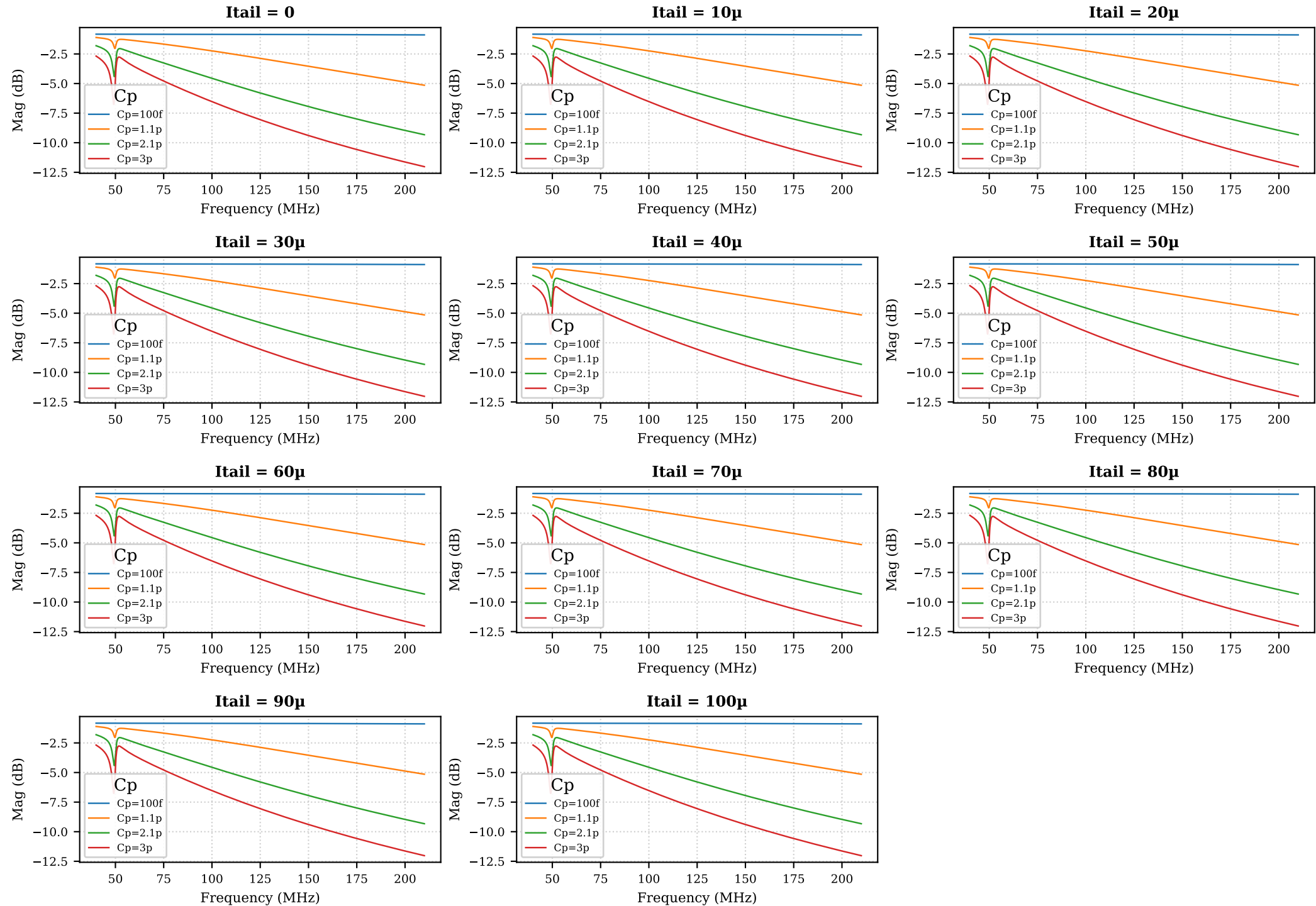
Cp = 2.1p



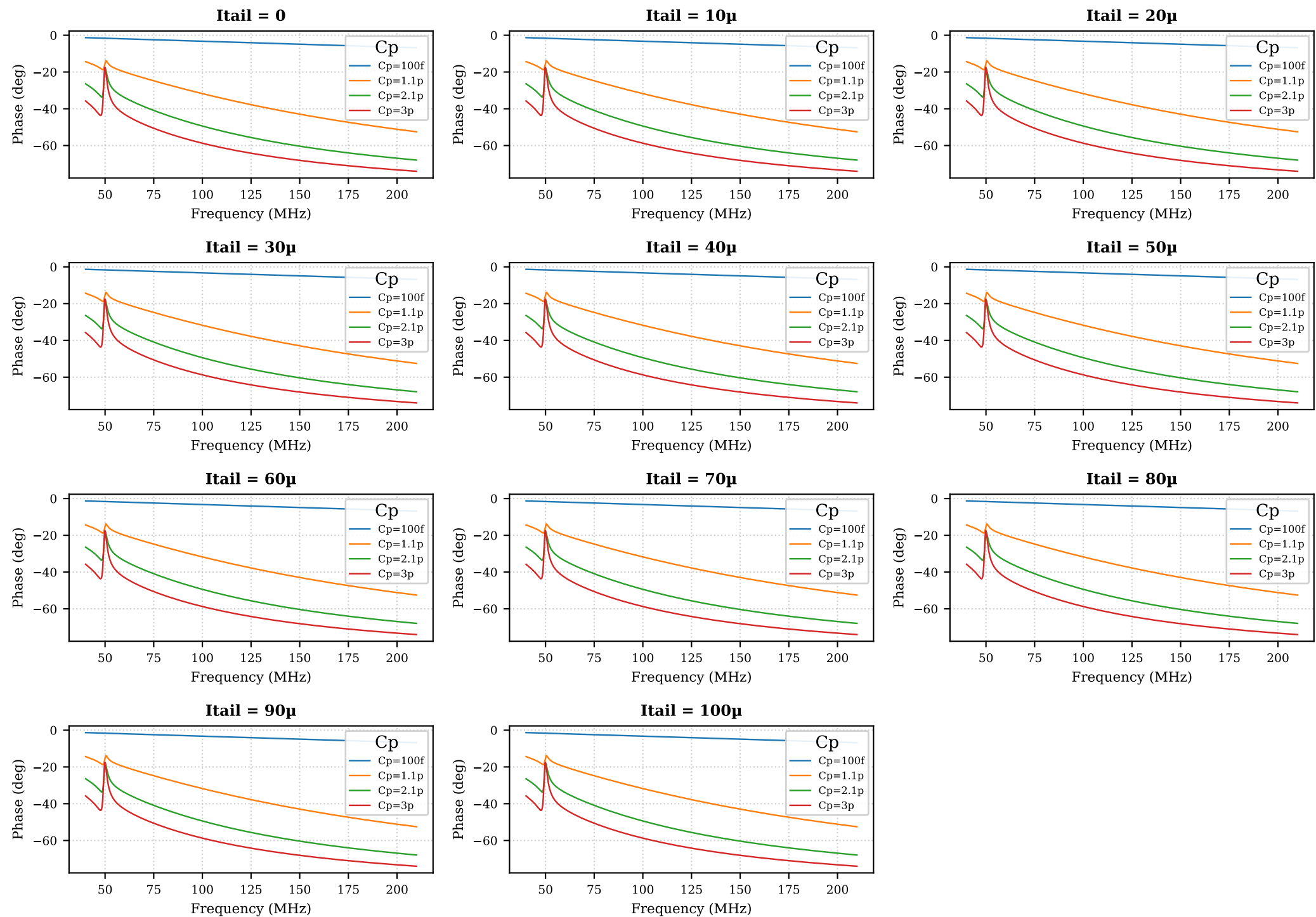
Cp = 3p



V(pas+) — Magnitude grouped by Itail (each subplot fixes Itail; legend = Cp)



V(pas+) — Phase grouped by Itail (each subplot fixes Itail; legend = Cp)



V(pas+) – Quality factor table (page 1/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(pas+)	100f	0	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	10μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	20μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	30μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	40μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	50μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	60μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	70μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	80μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	90μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	100f	100μ	210	-0.8597	0.03011			notch shallower than 3 d
V(pas+)	1.1p	0	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	10μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	20μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	30μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	40μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	50μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	60μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	70μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	80μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	90μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	100μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	2.1p	0	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	10μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	20μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	30μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	40μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	50μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	60μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	70μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	80μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	90μ	210	-6.558	2.767			notch shallower than 3 d

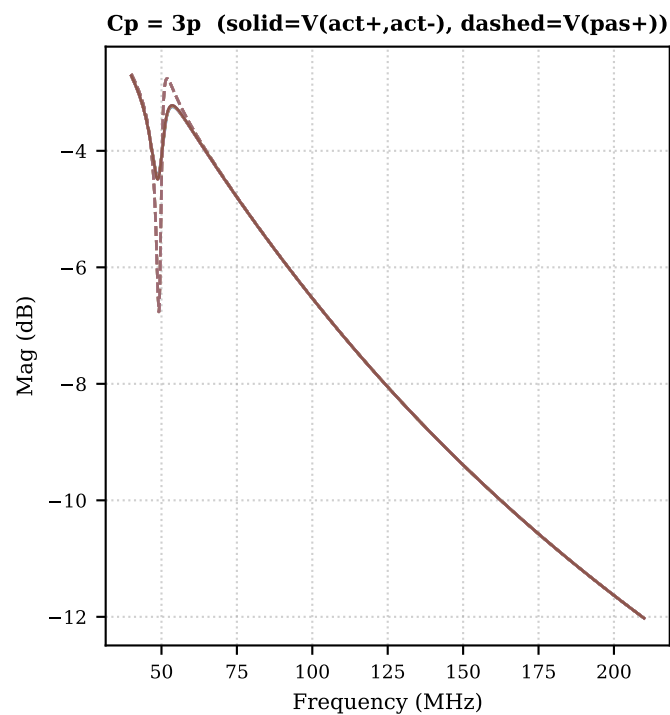
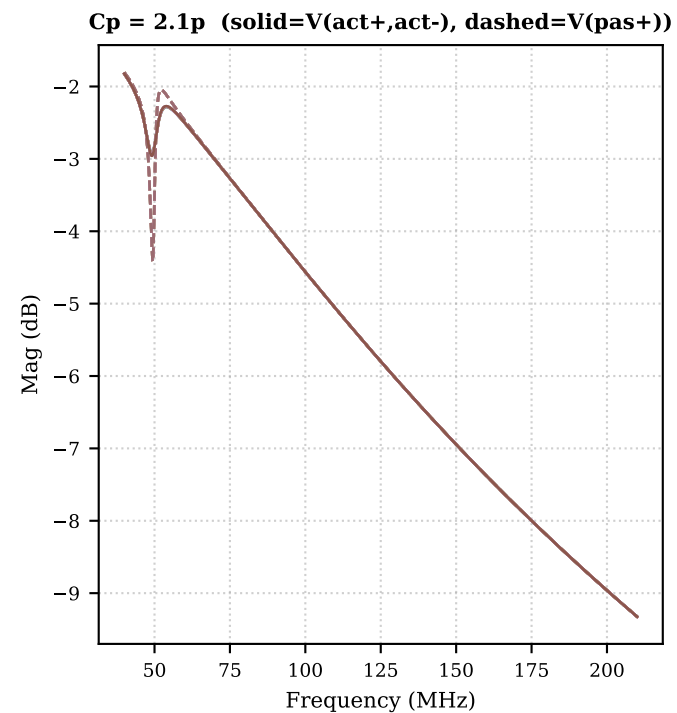
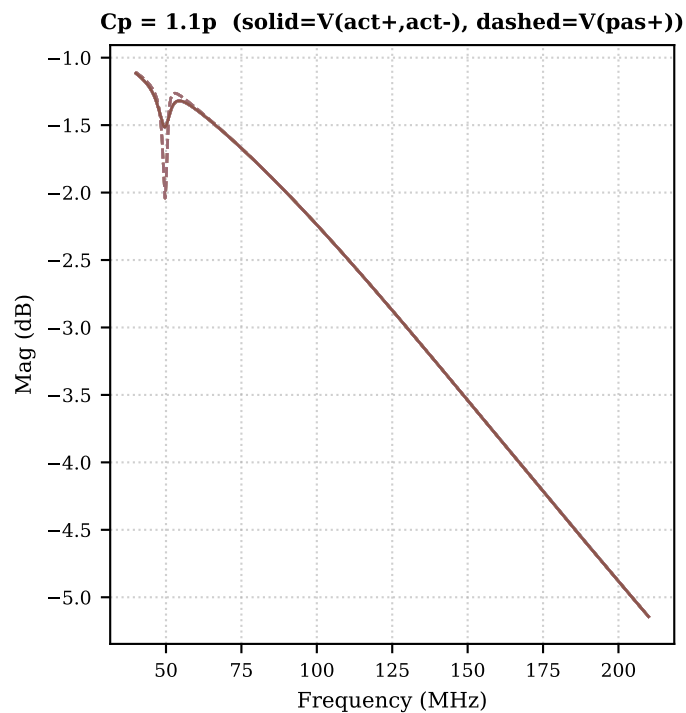
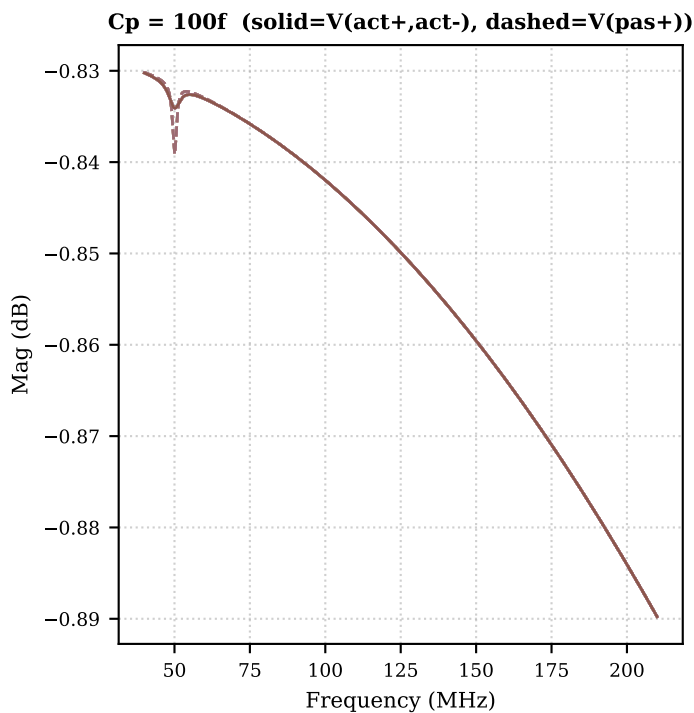
V(pas+) — Quality factor table (page 2/2)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(pas+)	2.1p	100μ	210	-6.558	2.767			notch shallower than 3 dB
V(pas+)	3p	0	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	10μ	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	20μ	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	30μ	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	40μ	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	50μ	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	60μ	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	70μ	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	80μ	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	90μ	210	-9.057	2.968			notch shallower than 3 dB
V(pas+)	3p	100μ	210	-9.057	2.968			notch shallower than 3 dB

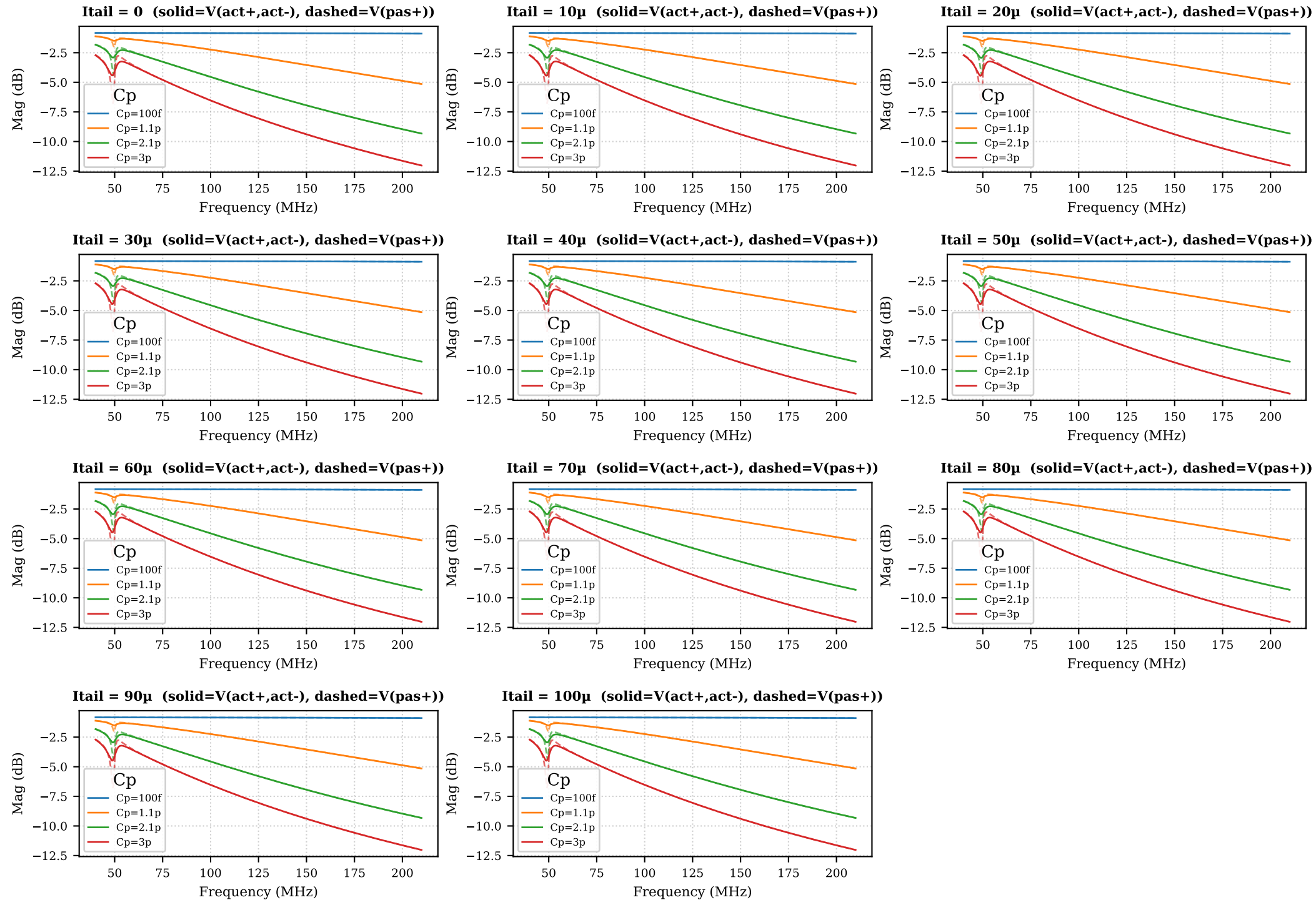
A. Active vs Passive comparison

Solid = $V(\text{act+}, \text{act-})$ Dashed = $V(\text{pas+})$

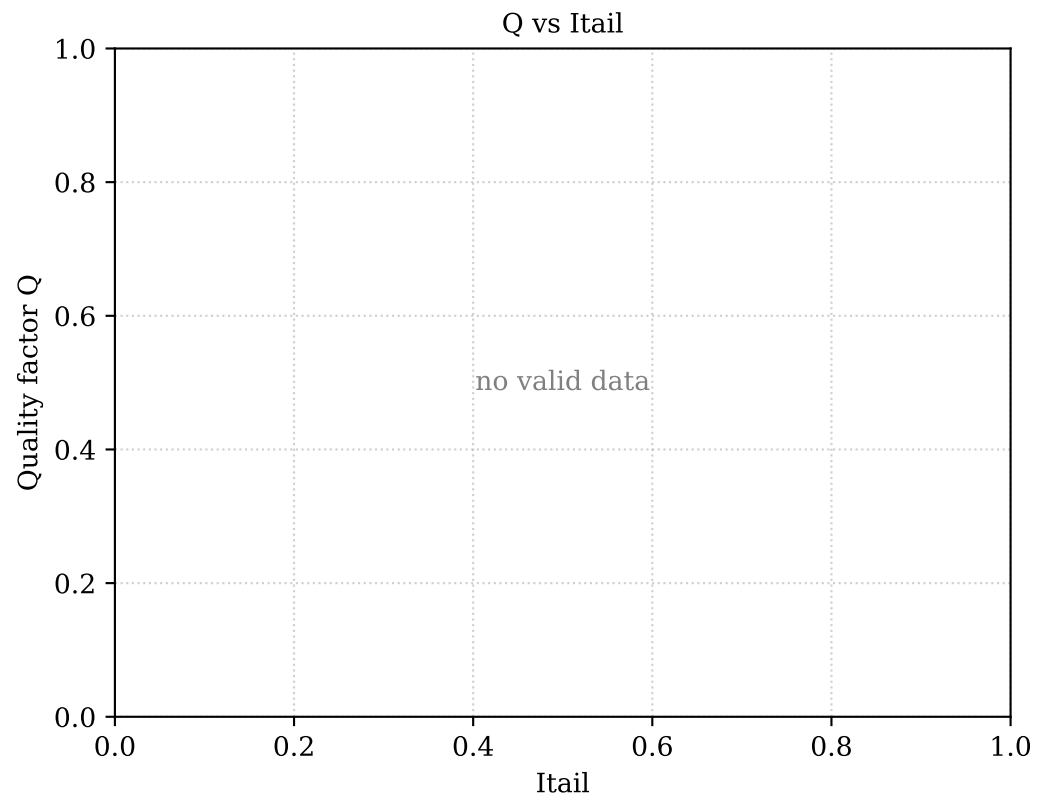
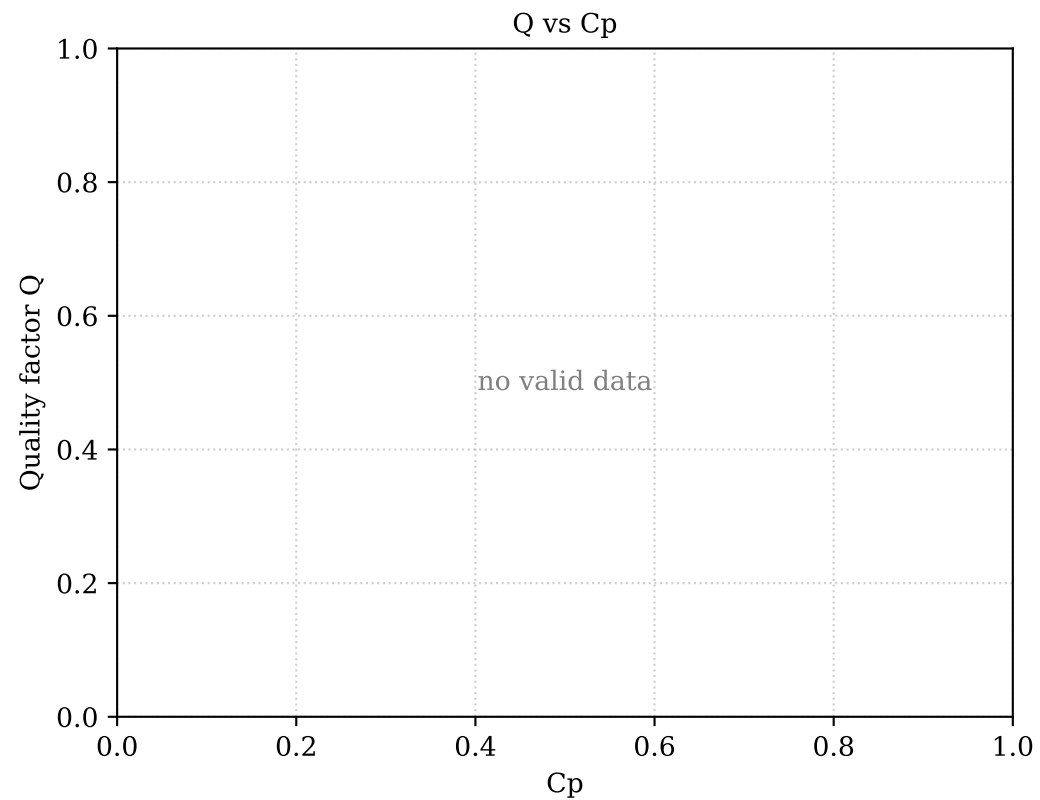
A. $V(\text{act+}, \text{act-})$ vs $V(\text{pas+})$ — grouped by C_p (legend = Itail)



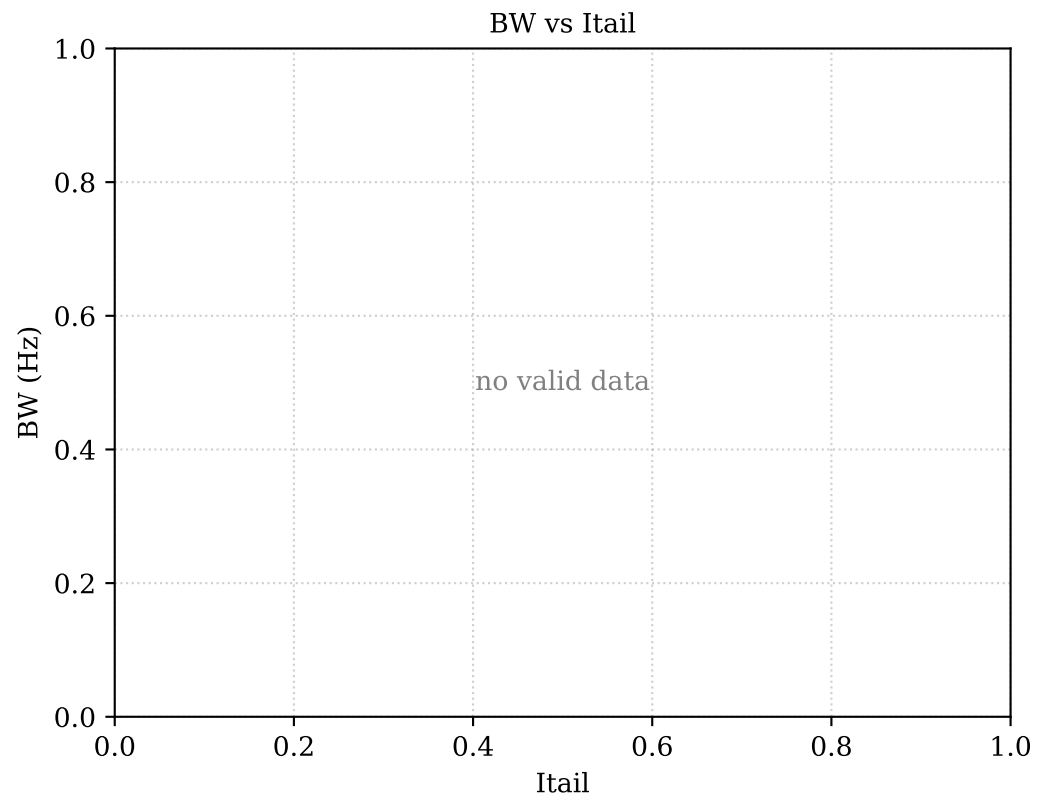
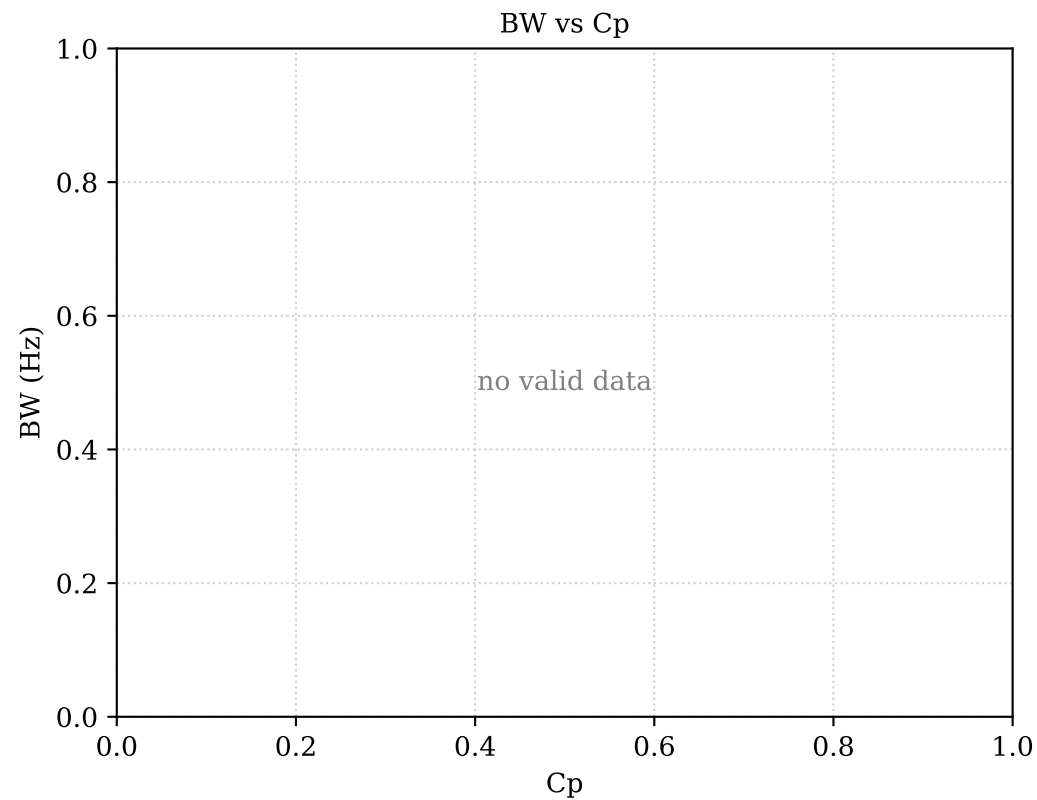
A. V(act+,act-) vs V(pas+) — grouped by Itail (legend = Cp)



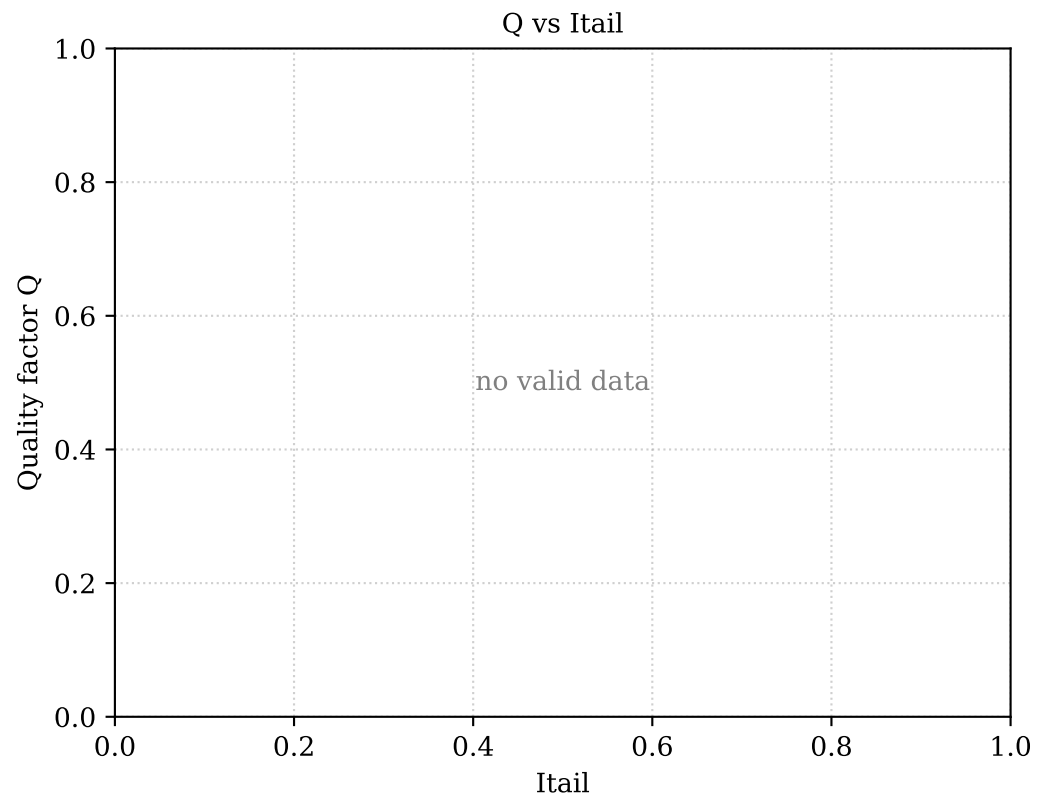
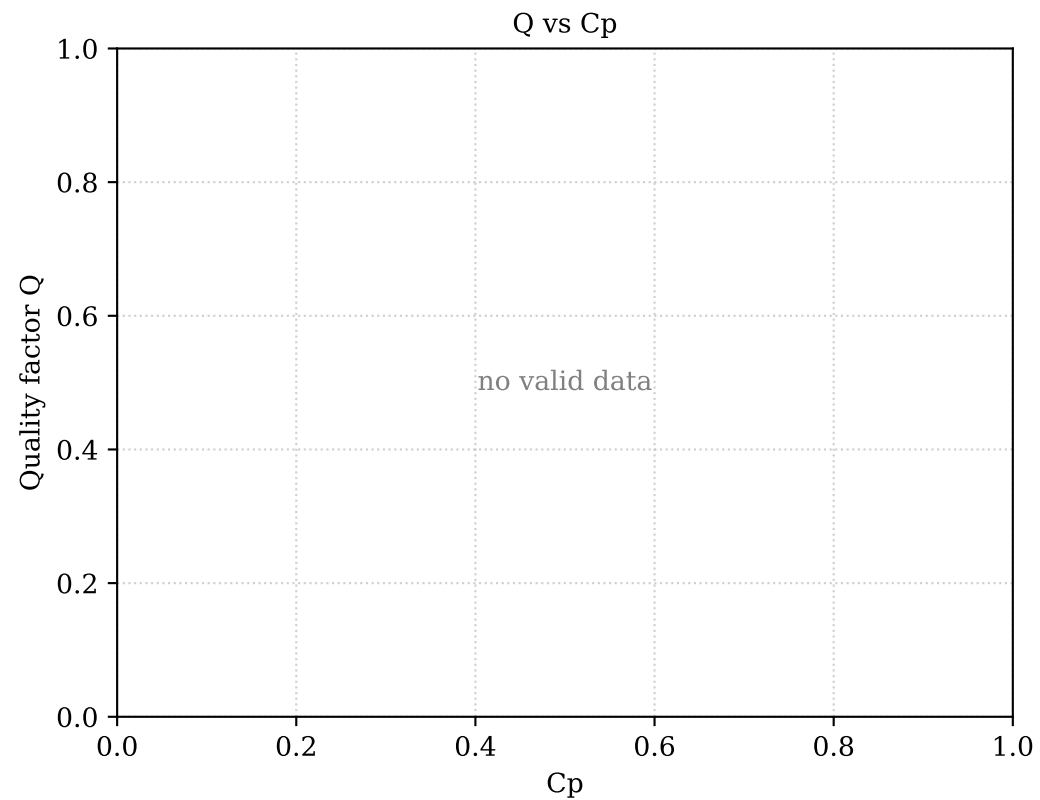
B. V(act+,act-) — Quality factor vs swept parameters



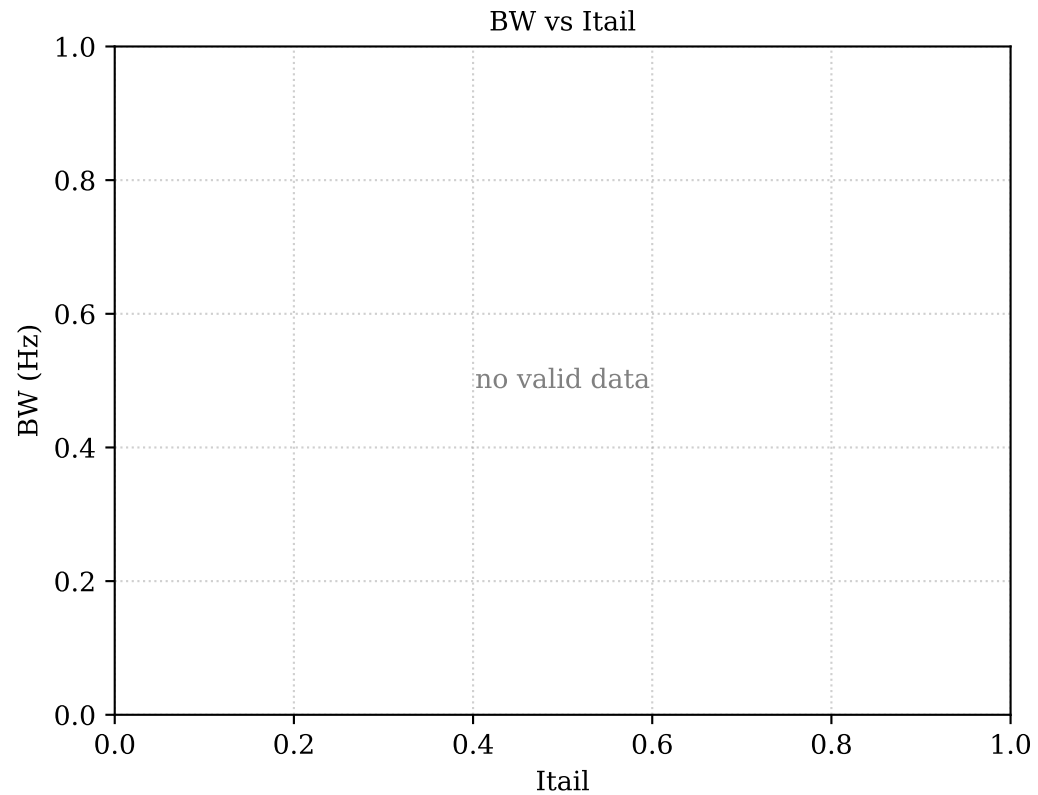
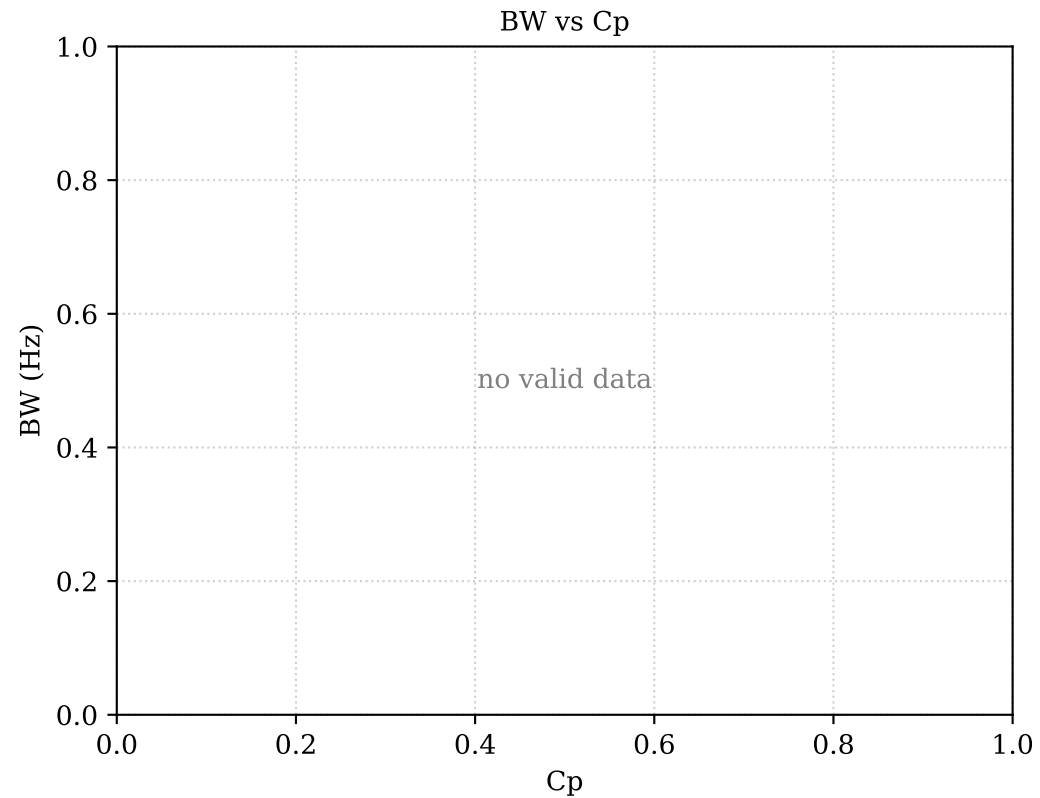
C. V(act+,act-) — Bandwidth (-3 dB) vs swept parameters



B. V(pas+) — Quality factor vs swept parameters



C. V(pas+) — Bandwidth (-3 dB) vs swept parameters



D. V(act+,act-) — Q heatmap (rows = Cp, columns = Itail)

	Itail=0	Itail=10μ	Itail=20μ	Itail=30μ	Itail=40μ	Itail=50μ	Itail=60μ	Itail=70μ	Itail=80μ	Itail=90μ	Itail=100μ
Cp=100f	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=1.1p	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=2.1p	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW
Cp=3p	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW	no BW

Red = BW could not be measured | Yellow = Q computed ($Q \leq 1400$) | Green = $Q > 1400$

E. Combined quality-factor table – all signals (page 1/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	100f	0	210	-0.8571	0.03268			notch shallower than 3 d
V(act+,act-)	100f	10μ	210	-0.8571	0.03266			notch shallower than 3 d
V(act+,act-)	100f	20μ	210	-0.8571	0.03265			notch shallower than 3 d
V(act+,act-)	100f	30μ	210	-0.8571	0.03264			notch shallower than 3 d
V(act+,act-)	100f	40μ	210	-0.8571	0.03263			notch shallower than 3 d
V(act+,act-)	100f	50μ	210	-0.8571	0.03263			notch shallower than 3 d
V(act+,act-)	100f	60μ	210	-0.8571	0.03262			notch shallower than 3 d
V(act+,act-)	100f	70μ	210	-0.8571	0.03262			notch shallower than 3 d
V(act+,act-)	100f	80μ	210	-0.8572	0.03262			notch shallower than 3 d
V(act+,act-)	100f	90μ	210	-0.8572	0.03261			notch shallower than 3 d
V(act+,act-)	100f	100μ	210	-0.8572	0.03261			notch shallower than 3 d
V(act+,act-)	1.1p	0	210	-3.099	2.045			notch shallower than 3 d
V(act+,act-)	1.1p	10μ	210	-3.101	2.043			notch shallower than 3 d
V(act+,act-)	1.1p	20μ	210	-3.102	2.042			notch shallower than 3 d
V(act+,act-)	1.1p	30μ	210	-3.103	2.041			notch shallower than 3 d
V(act+,act-)	1.1p	40μ	210	-3.104	2.04			notch shallower than 3 d
V(act+,act-)	1.1p	50μ	210	-3.104	2.04			notch shallower than 3 d
V(act+,act-)	1.1p	60μ	210	-3.105	2.039			notch shallower than 3 d
V(act+,act-)	1.1p	70μ	210	-3.105	2.039			notch shallower than 3 d
V(act+,act-)	1.1p	80μ	210	-3.106	2.038			notch shallower than 3 d
V(act+,act-)	1.1p	90μ	210	-3.106	2.038			notch shallower than 3 d
V(act+,act-)	1.1p	100μ	210	-3.106	2.037			notch shallower than 3 d
V(act+,act-)	2.1p	0	210	-5.809	3.516			right -3dB not crossed
V(act+,act-)	2.1p	10μ	210	-5.816	3.509			right -3dB not crossed
V(act+,act-)	2.1p	20μ	210	-5.818	3.507			right -3dB not crossed
V(act+,act-)	2.1p	30μ	210	-5.82	3.504			right -3dB not crossed
V(act+,act-)	2.1p	40μ	210	-5.822	3.503			right -3dB not crossed
V(act+,act-)	2.1p	50μ	210	-5.824	3.501			right -3dB not crossed
V(act+,act-)	2.1p	60μ	210	-5.825	3.5			right -3dB not crossed
V(act+,act-)	2.1p	70μ	210	-5.827	3.498			right -3dB not crossed
V(act+,act-)	2.1p	80μ	210	-5.828	3.497			right -3dB not crossed
V(act+,act-)	2.1p	90μ	210	-5.829	3.496			right -3dB not crossed

E. Combined quality-factor table – all signals (page 2/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(act+,act-)	2.1p	100μ	210	-5.83	3.495			right -3dB not crossed
V(act+,act-)	3p	0	210	-7.887	4.137			right -3dB not crossed
V(act+,act-)	3p	10μ	210	-7.897	4.127			right -3dB not crossed
V(act+,act-)	3p	20μ	210	-7.902	4.123			right -3dB not crossed
V(act+,act-)	3p	30μ	210	-7.905	4.119			right -3dB not crossed
V(act+,act-)	3p	40μ	210	-7.908	4.116			right -3dB not crossed
V(act+,act-)	3p	50μ	210	-7.91	4.114			right -3dB not crossed
V(act+,act-)	3p	60μ	210	-7.913	4.111			right -3dB not crossed
V(act+,act-)	3p	70μ	210	-7.915	4.109			right -3dB not crossed
V(act+,act-)	3p	80μ	210	-7.917	4.107			right -3dB not crossed
V(act+,act-)	3p	90μ	210	-7.919	4.105			right -3dB not crossed
V(act+,act-)	3p	100μ	210	-7.921	4.103			right -3dB not crossed
V(pas+)	100f	0	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	10μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	20μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	30μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	40μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	50μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	60μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	70μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	80μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	90μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	100f	100μ	210	-0.8597	0.03011			notch shallower than 3 dB
V(pas+)	1.1p	0	210	-3.369	1.775			notch shallower than 3 dB
V(pas+)	1.1p	10μ	210	-3.369	1.775			notch shallower than 3 dB
V(pas+)	1.1p	20μ	210	-3.369	1.775			notch shallower than 3 dB
V(pas+)	1.1p	30μ	210	-3.369	1.775			notch shallower than 3 dB
V(pas+)	1.1p	40μ	210	-3.369	1.775			notch shallower than 3 dB
V(pas+)	1.1p	50μ	210	-3.369	1.775			notch shallower than 3 dB
V(pas+)	1.1p	60μ	210	-3.369	1.775			notch shallower than 3 dB
V(pas+)	1.1p	70μ	210	-3.369	1.775			notch shallower than 3 dB
V(pas+)	1.1p	80μ	210	-3.369	1.775			notch shallower than 3 dB

E. Combined quality-factor table — all signals (page 3/3)

Signal	Cp	Itail	f0 (MHz)	Baseline (dB)	Depth (dB)	BW (Hz)	Q	Note
V(pas+)	1.1p	90μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	1.1p	100μ	210	-3.369	1.775			notch shallower than 3 d
V(pas+)	2.1p	0	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	10μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	20μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	30μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	40μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	50μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	60μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	70μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	80μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	90μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	2.1p	100μ	210	-6.558	2.767			notch shallower than 3 d
V(pas+)	3p	0	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	10μ	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	20μ	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	30μ	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	40μ	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	50μ	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	60μ	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	70μ	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	80μ	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	90μ	210	-9.057	2.968			notch shallower than 3 d
V(pas+)	3p	100μ	210	-9.057	2.968			notch shallower than 3 d